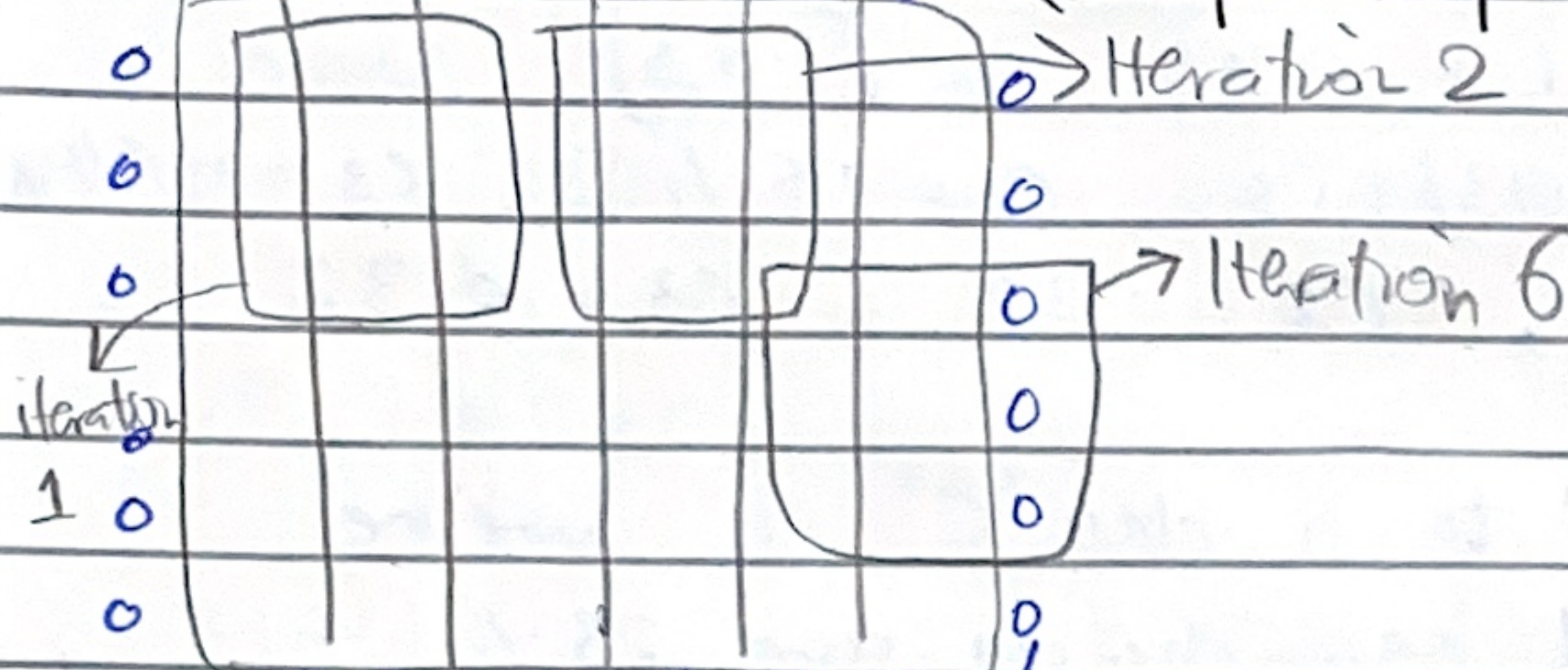


LECTURE-23

Wednesday
29/4/26.

★ Padding need only if $\text{slide} > 1$ as we jump outta image boundy.

If $\text{Slide} = 2$ (in prev page's ques) in 6×6 matrix:



→ In case of padding; the center of filter will be on top left of Image (& rest on outer padding).

This is padding and we did so if $\text{slide} = 2$, the 3rd column of filter should not ~~overflow~~ overflow out of Image

⇒ Padding: It's to add zeros on all sides of Image

→ Valid: No padding.

→ Same: Original Image's dimension will be given in output Feature map like in this case 6×6 will be in output as after padding we've 8×8 (zeros on borders included)

- Filter

$$\text{2D Dimension of Output feature map} = \frac{\text{Image} + 2(\text{Padding})}{\text{slide}} + 1$$

e.g: $D = \frac{6 + 2(1)}{1} + 1 = 6 \times 6$ ✓ (same dimension as Image after convolution)

Q. Image 128×128 . Filter: 5×5 , $\text{slide} = 1$

(a) Padding = valid; 0

$$D = \frac{128 - 5 + 2(0)}{1} + 1$$

$$D = 124 \times 124$$

(b) Padding = Same

Need to choose padding so it gives 128×128 matrix in output as image is also 128×128 .

That padding will be chosen on run time

For Rectangular Image: Image 50×120
Rectangular Filter Filter 4×3

$I_H = 50$, $I_W = 120$ } $\therefore$ therefore now we'll have
 $F_H = 4$, $F_W = 3$ } Output dimension Key 2 formulas;
one for height & one for width

• $D_H = \frac{I_H - F_H + 2(P)}{S} + 1$ (output dimension's height)

• $D_W = \frac{I_W - F_W + 2(P)}{S} + 1$ (output dimension's width).

Image Types:

- Binary (0, 1): Black & white only. 128×128
- Grey Scale (0...1): Towards 0 is blackish & towards 1 is whitish $128 \times 128 \times 1$
- RGB $128 \times 128 \times 3$.

- When image comes in RGB; it comes in in a 2D Red matrix, 2D Blue and a 2D green matrix (not stacked)
- The filter will now be $3 \times 3 \times 3$ since three coloured matrices. These are just three layer of one single filter with diff. weights in each layer.

Image $\xrightarrow{\text{output feature (slide=1) map}}$

R (4x4) $\rightarrow$ 01: 2x2

G (4x4) $\rightarrow$ 02: 2x2

B (4x4) $\rightarrow$ 03: 2x2

channel 1 of filter for R

Filters (bias = 1) channel 2 of filter for G

C1 3x3

C2 3x3

C3 3x3

3x3x3 for 3 colors

$$\begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} + \text{bias } 1 = \begin{bmatrix} 2+1+0+1 & 0+0+0+1 \\ 1+1+2+1 & 3+0+3+1 \end{bmatrix}$$

now this is output feature map from single filter $\therefore 2 \times 2 \times 1$.

$$= \begin{bmatrix} 4 & 2 \\ 5 & 7 \end{bmatrix}$$

* Each layer of filter will be sliding over each coloured matrix (like: C1 goes over R, C2 on G ...) & then Net gives us 01, 02, 03 output feature maps.

Default $\rightarrow$ Slide = 1, Padding = valid.

Layers	No. of Filters	Filter size	Slide	Padding	Output Dimensions	No. of Parameters
Input Im.					A 128x128x3	
Conv. 1	D 64	B 5x5	C 1	Valid	1 124x124x64	9 4864
Pool 1		2	2		2 62x62x64	
Conv. 2	E 128	5x5	1	Same	3 62x62x128	10 328 204298
Pool 2		2	2		4 31x31x128	
Flatten					5 123008	
Dense 1	50				6 1x50	11 6150450
Dense 2	20				7 1x20	12 1020
Output	10				8 1x10	13 210

converting into 1D array for giving it to ANN now

~~Output~~

$$\Sigma = 6156672$$

① Output Dimension = Image - Filter + 1

from Input row in cell (A) $\xleftarrow{\text{Slide cell (B)}}$ $= 128 - 5 + 1 = 124 \times 124 \times 64$

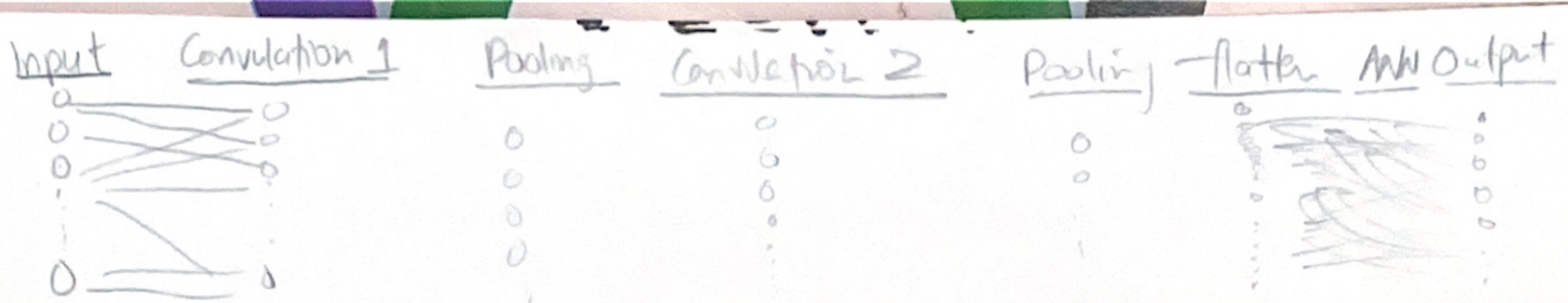
1 $\rightarrow$ cell (C)

no. of filters = depth: from cell (D)

② Half the Height & width from ① as that is the input of Pool 1.

124x124 $\rightarrow$ 62x62x64 $\rightarrow$ Same no. of filters as in ① $\therefore$ they won't get halved.

if decimal answer; then do floor func.



③ Since padding = same $\therefore$ The output dimension will be same as the input dimension (only height and width) which is Pool 1's output for conv. 1 Input $\therefore 62 \times 62$.
And since we've 128 filters in cell E $\therefore \times 128$ depth.

④ Halving the height & width of Input i.e.: conv's 2 output as pooling reduces size whereas depth (128) stays same as they're just no. of filters.

⑤ Simply multiply the 3 values (dimension of 3D array) in input i.e.: $31 \times 31 \times 128$ to convert it into 1D

⑥ ⑦ ⑧ These are just hidden and output layers of ANN/perceptron and 50, 20, 10 are no. of neurons in each layer.

⑨ No. of Parameter = $\left[(F_H * F_W * F_D) + 1 \right] * (\text{No. of Filters})$
 $\therefore$ since depth of Input in cell (A) is 3

$(5 \times 5 \times 3) + 1 = 76$ (for one filter we've these many

$$76 * 64 = 4864$$

$\rightarrow$ No. of filters

⑩ No. of Para. = $\left[(5 \times 5 \times 64) + 1 \right] \times 128$
 $\text{input from 5 for dense} = 1204928$

⑪ $(123008 \times 50) + 50 = 6150450$

$\rightarrow$ no. of neuron $\rightarrow$ each neuron has 1 bias $\therefore$ 50 total biases

⑫ ⑬ $(50 \times 20) + 20 = 1020$ & $(20 \times 10) + 10 = 210$

⑭ Sum of all No. of Parameters

$\Rightarrow$ If asked to change from RGB to grey scale:

We'll only change the output dim. of first row of Input Img.

by $128 \times 128 \times 1$ from $(128 \times 128 \times 3)$ $\therefore$ only No. of parameter for Conv. 1 changes rest all will have same effect with depth = 1 (as one channel of filter only).